ELEC40006 Electronics Design Project

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1 Mechanical and Hardware Integration

The mechanical and hardware architecture of the EEELunarRover was developed as an integrated platform rather than as a passive enclosure for the electronics. The project brief requires the rover to survey an artificial lunar surface, classify rocks using radio, ultrasonic, infrared, and magnetic signals, remain cost and weight effective, negotiate uncrossable obstacles, and operate reliably during the final arena demonstration. Mechanical design therefore affects the whole system: it determines whether the rover can reach each rock, whether the sensors can be held at a repeatable standoff, whether the motor system can manoeuvre in the available space, and whether the final rover stays below the 750 g weight-sensitive-zone limit.

The arena constraints were converted into mechanical targets before component selection. Rocks are spaced at least 500 mm apart on a grey rubber surface with uncrossable obstacles, so the rover must manoeuvre through a constrained environment rather than simply drive in a straight line. The sensing subsystems also impose a front-end geometry requirement: the radio coil, ultrasonic transducer, infrared detector, and magnetometer must be able to approach a target closely enough for repeatable readings without the chassis or wheels contacting the rock first. The adopted packaging target was therefore to place the front sensing geometry within approximately 50 mm of a rock while keeping the main chassis clear.

1.1 Derived Mechanical and Hardware Requirements

Table 1 gives the derived requirements used to drive the mechanical, power, and motor-control design. The requirements are deliberately phrased as verifiable statements so that the final evaluation can test the same claims made during the design process.

ID	Requirement	Verification method	Status
MH1	Total rover mass, including chassis, batteries, fasteners, wiring, motor drivers, and electronics, must remain below 750 g.	Final mass measurement using calibrated scale with batteries installed.	[TBD]
MH2	Rover must manoeuvre between rocks and around uncrossable arena obstacles.	Representative arena driving trial with rocks and barriers.	Validated
MH3	Rover must make small position corrections near a target rock without needing a large turning radius.	Low-speed forward, lateral, rotational, and final-alignment tests.	Validated
MH4	Sensor mounting must keep the radio coil, ultrasonic transducer, infrared detector, and magnetometer mechanically stable during motion.	Visual inspection, repeated approach tests, and sensor-repeatability checks before and after driving.	[TBD]
MH5	Motor switching and power transients must not reset the microcontroller or invalidate sensor readings.	Logic-rail, motor-rail, and sensor-output measurements during startup, reversal, and driving.	Validated
MH6	Motor-control hardware must provide independent bidirectional control of all four wheels.	Direction and PWM tests for each motor channel.	Validated
MH7	Mechanical layout must reserve sufficient space for the PCB or stripboard, battery holder, Adafruit board, WiFi shield, motor drivers, and sensors.	CAD assembly check and final physical integration check.	Validated

Table 1: Derived mechanical and hardware requirements used to guide rover integration.

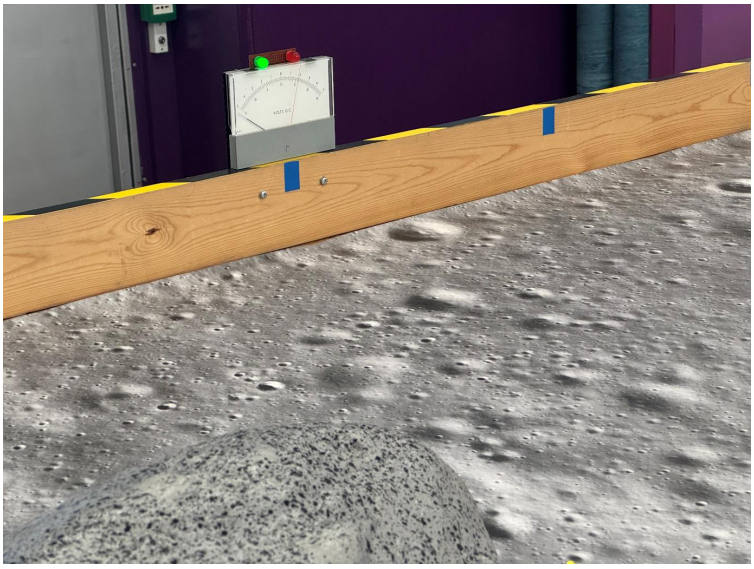


Figure 1: Arena constraints, including rock obstacles and the weight-sensitive zone enforcing the 750 g mass limit.

1.2 Wheel and Drive-System Selection

1.2.1 Drive-system alternatives

The drive system was selected for manoeuvrability, controllability near rocks, packaging efficiency, and compatibility with the mass limit. A steering system that couples translation and rotation would make the final approach slower, because sensor alignment would require repeated forward–backward corrections and rotations. A holonomic drivetrain is therefore attractive despite its additional motor-control complexity.

Drive option	**Advantages**	**Reason for rejection or selection**
Two-wheel differential drive	Mechanically simple, low mass, easy to control.	Rejected because it cannot translate sideways; close-range target alignment would need repeated rotations and reversals.
Ackermann steering	Stable for forward motion and familiar for vehicle platforms.	Rejected because it requires a finite turning radius and cannot rotate in place or strafe around obstacles.
Skid or tank drive	Can rotate approximately in place and is mechanically robust.	Rejected because turning relies on wheel slip, wasting power and reducing repeatability on the rubber arena surface.
Three-wheel omni drive	Holonomic motion with fewer wheels and motors than four-wheel mecanum.	Rejected because triangular packaging is less suitable for the rectangular battery, PCB, sensor board, and motor-driver layout; front sensor geometry is also less direct.
Four-wheel mecanum drive	Holonomic motion, rectangular packaging, independent wheel control, and direct lateral correction near rocks.	Selected despite higher mass and wiring complexity because it best addresses the dominant mechanical risk: slow or inaccurate sensor alignment near rocks.

Table 2: Comparison of candidate rover drive systems.

The final design uses four independently driven 60 mm mecanum wheels. This enables forward/backward translation, lateral translation, in-place rotation, and combined translation with rotation. The main design benefit is not maximum speed but low-speed pose correction: after detecting a rock, the operator can adjust the rover laterally without losing the sensor orientation.

1.2.2 First-order force comparison

A simplified force comparison was used to support the drivetrain choice. The model is not intended to predict exact traction because it neglects roller friction, slip, load distribution, motor mismatch, and drivetrain losses. It is still useful because the same idealised assumptions are applied to both candidate holonomic layouts, allowing a first-order comparison of available force directions.

Assume each motor produces torque τ, each wheel has radius r, and each wheel produces ideal ground force T, where

$$T = \frac{\tau}{r}.$$

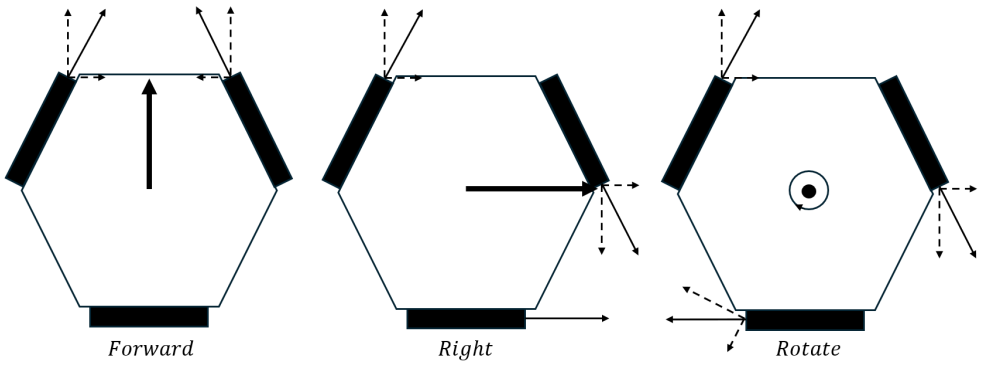


Figure 2: Top-view force-component analysis of a three-wheel omni-wheel drive.

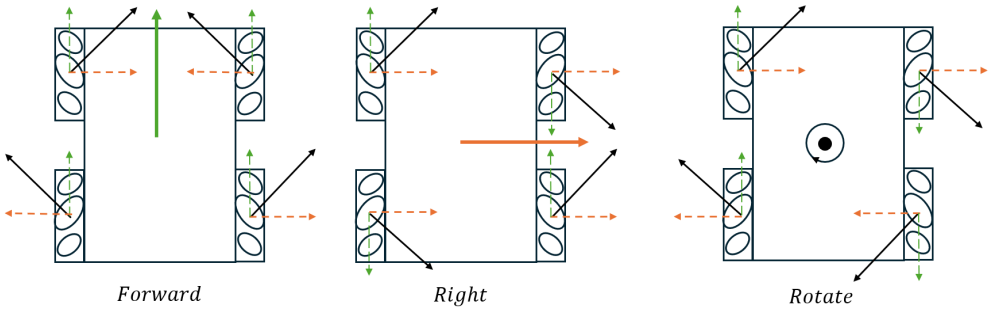


Figure 3: Top-view force-component analysis of a four-wheel mecanum-wheel drive.

For forward motion, the ideal resultant forces are

$$F_{\mathrm{O,forward}} = 2T\sin(60^\circ) = \sqrt{3}T \approx 1.73T,$$

$$F_{\mathrm{M,forward}} = 4T\cos(45^\circ) = 2\sqrt{2}T \approx 2.83T.$$

For sideways motion,

$$F_{\mathrm{O,lateral}} = 2T\cos(60^\circ) + T = 2T,$$

$$F_{\mathrm{M,lateral}} = 4T\cos(45^\circ) = 2\sqrt{2}T \approx 2.83T.$$

For rotation, where s is the effective distance from the chassis centre to the wheel force line,

$$M_{\mathrm{O}} = 3sT, \qquad M_{\mathrm{M}} = 4\sqrt{2}sT \approx 5.66sT.$$

Under this model, the four-wheel mecanum configuration provides greater theoretical forward force, lateral force, and yaw moment than the three-wheel omni layout. The result was not used in isolation: mecanum drive requires an extra motor and wheel, two dual H-bridge drivers, additional wiring, more PWM/direction pins, and a larger mass allocation. It was nevertheless selected because the added complexity directly reduces the most important operational risk, namely poor close-range alignment between the sensors and the target rock.

1.2.3 Wheel diameter selection

The 60 mm wheel diameter was selected as a compromise between ground clearance, torque demand, controllability, and speed. For fixed motor torque, the available tangential ground force decreases as wheel radius increases:

$$F = \frac{\tau}{r}.$$

A larger wheel would improve clearance and linear speed but reduce available tractive force and make fine low-speed movements harder. A smaller wheel would increase ground force but reduce clearance, making the rover more vulnerable to surface irregularities, cable drag, and small obstacles. The selected diameter gives moderate clearance while keeping torque demand practical for the selected TT motors.

Test criterion	**Target**	**Measured**	**Result**
Minimum controllable forward displacement (0.1 s)	10 mm	5.5 mm at PWM 100	Pass
Minimum controllable lateral displacement (0.1 s)	10 mm	5.0 mm at PWM 100	Pass
Travel speed over 500 mm at PWM 255	200 mm/s	250 mm/s	Pass
Time to align sensor face with target rock	[TBD: target s]	[TBD: measured s]	[TBD]
Current draw during straight driving	< 0.6 A	0.123 A	Pass
Current draw during lateral motion or turning	< 0.6 A	0.121 A	Pass

Table 3: Experimental validation criteria for the 60 mm wheel selection. All Reading Taken as Average of 10.

Most of the validation criteria was well over expectation, meaning that 60 mm mecanum wheel was a suitable choice.

1.2.4 Mecanum wheel manufacture and iteration

The objective was to manufacture four functional mecanum wheels using available laboratory processes and no additional budget. The wheel system needed low-friction roller motion, adequate structural stiffness, tyre grip, and repeatable assembly. The design was decomposed into ten angled rollers per wheel, roller supports, shafts, bearing features, and a three-part cap assembly so that individual failure modes could be isolated before integration.

The CAD files were reviewed in eDrawings and prepared for slicing in Ultimaker Cura after STL conversion. The initial assumption was that all parts, including shafts and bearing features, could be printed. Dimensional checks were then used to assess assembly feasibility. The roller

support had an outer diameter of 5.4 mm, a middle diameter of 4.2 mm, and an inner bore of 2.4 mm. This was intended to accept a bearing feature with a 4.16 mm middle diameter while leaving clearance for a 1.8 mm shaft. The TPU roller had a 4.86 mm inner diameter, deliberately smaller than the 5.4 mm support diameter so that material elasticity would create an interference fit and improve traction.

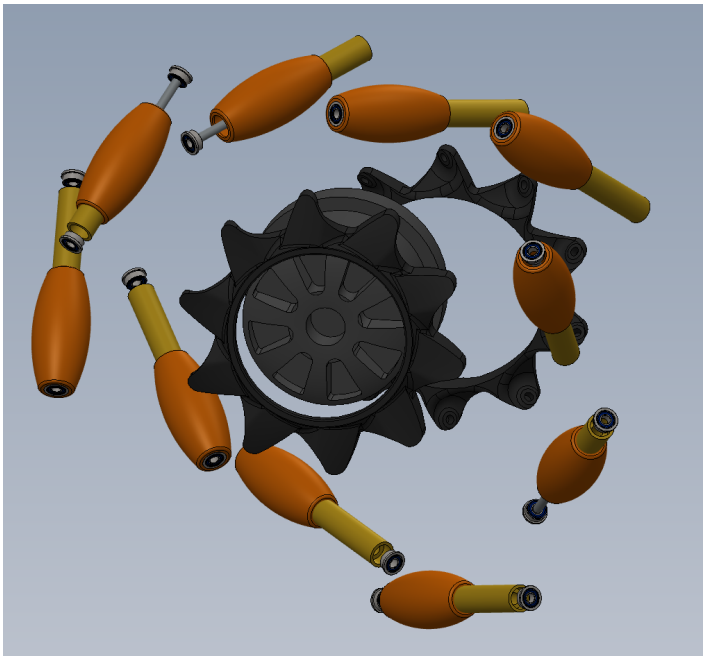


Figure 4: Exploded CAD view showing the roller, shaft, bearing, and cap interfaces.

Part	**Material**	**Infill**	**Purpose**
Roller	TPU	25 %, gyroid	Grip and compliance.
Roller support	PETG	50 %, cubic	Structural roller core.
Shaft	PETG	100 %	Roller axle in the initial design.
Caps	PETG	50 %, cubic	Main wheel frame.
Bearing feature	PLA	100 %	Low-friction support in the initial design.

Table 4: Initial manufacturing plan for the custom mecanum wheels.

The first print trial tested one roller and one cap. The TPU roller and PETG support assembled successfully, confirming the interference-fit concept. However, the printed bearing and shaft concepts failed at the available print resolution. The nominal shaft diameter was 1.8 mm, but the laboratory nozzle was 0.4 mm and deposited an effective bead of approximately 0.6 mm; the printed shaft therefore became an uneven stack and snapped easily. Small cap and support details were also poorly resolved around holes and interfaces.



Figure 5: Loss of definition in small printed features during the first wheel-manufacturing iteration.

The redesign removed the printed bearing feature and replaced the printed shaft with a metal rod. After comparing wires, nails, and workshop stock, a rivet mandrel was selected because its measured dimensions were close to the required axle size: approximately 1.9 mm diameter and 29 mm length. The cutting process was also changed after the first shaft was too short: the rivet pin was cut first, then the mandrel was trimmed and filed to length. This produced shafts long enough to be fixed into both caps while allowing each roller to rotate freely.

Final wheel assembly used printed TPU rollers, printed supports and caps, and metal mandrel shafts. The shafts were glued into one side of the cap, rollers were placed onto the shafts, and the opposite cap was fitted and glued to lock the structure. The final wheel preserved the intended roller orientation and achieved independent roller motion.



Figure 6: Completed wheel after replacing the printed shaft and bearing concept with metal mandrel shafts.

Failure mode	Evidence	Design change	Outcome
Printed shaft fracture	1.8 mm shaft was too fine relative to 0.4 mm nozzle and 0.6 mm effective bead.	Replace with 1.9 mm rivet mandrel.	Passed assembly and rotation check.
Poor bearing definition	Printed bearing geometry did not give useful rolling behaviour at this scale.	Remove bearing and rely on metal shaft/roller clearance.	Simpler, more repeatable assembly.
Short first shaft cut	First mandrel piece was too short to be retained in both caps.	Cut rivet pin first, then trim and file mandrel.	Repeatable shaft length for wheel assembly.
Roller grip uncertainty	Roller needed traction without excessive stiffness.	TPU roller with interference fit over support.	Snug fit and visible traction.

Table 5: Wheel-manufacturing failure modes, evidence, design response, and outcome.

At subsystem level, the final wheel design passed the practical checks available before full arena testing: free roller rotation, adequate structural integrity, and visible traction from the TPU rollers. The strongest quantitative evidence was dimensional: a 1.8 mm printed shaft was not feasible with the printer resolution used, whereas the 1.9 mm rivet mandrel matched the 2.4 mm support bore with practical clearance. At system level, the wheels were later integrated with the chassis and operated smoothly during first movement tests. The final report should include the linked test video and a measured displacement-error table once repeated motion trials have been completed.

1.3 Power Distribution and Motor Control

1.3.1 Motor-drive requirement

The four 6 V DC motors cannot be driven directly from the Adafruit microcontroller pins. The board uses a Microchip SAM D21/DA1 microcontroller, whose GPIO pins are intended for logic-level signalling rather than motor power. Each TT motor requires approximately 100 mA during normal operation, with higher transient current during startup, rapid direction changes, and blocked-wheel conditions. Direct connection would exceed safe I/O current capability, risk permanent microcontroller damage, and still would not provide bidirectional control.

Table 40-14. Normal I/O Pin Characteristics

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Units
V_{IL}	Input low-level voltage	VDD=2.7V-3.63V	-	-	0.3*VDD	V
V_{IH}	Input high-level voltage	VDD=2.7V-3.63V	0.55*VDD	-	-	
V_{OL}	Output low-level voltage	VDD>2.7V, IOL max.	-	0.1*VDD	0.2*VDD	
V_{OH}	Output high-level voltage	VDD>2.7V, IOH max.	0.8*VDD	0.9*VDD	-	
I_{OL}	Output low-level current	VDD=2.7V-3V,	-	-	-	mA
		PORT.PINCFG.DRVSTR=0	-	-	1	
		VDD=3V-3.63V,	-	-	-	
		PORT.PINCFG.DRVSTR=0	-	-	2.5	
		VDD=2.7V-3V,	-	-	-	
		PORT.PINCFG.DRVSTR=1	-	-	3	
		VDD = 3V-3.63V,	-	-	-	
		PORT.PINCFG.DRVSTR=1	-	-	10	
I_{OH}	Output high-level current	VDD=2.7V-3V,	-	-	-	
		PORT.PINCFG.DRVSTR=0	-	-	0.7	
		VDD=3V-3.63V,	-	-	-	
		PORT.PINCFG.DRVSTR=0	-	-	2	
		VDD=2.7V-3V,	-	-	-	
		PORT.PINCFG.DRVSTR=1	-	-	2	
		VDD=3V-3.63V,	-	-	-	
		PORT.PINCFG.DRVSTR=1	-	-	7	

Figure 7: Microchip SAM D21/DA1 I/O pin current limits, showing that GPIO pins are unsuitable for directly driving the DC motors.

To satisfy MH6, two Toshiba TC78H620FNG dual H-bridge motor-driver ICs were used. Each IC contains two full H-bridge channels, so two drivers provide independent bidirectional control of all four wheels. The motor current is supplied by the H-bridge outputs, while the microcontroller only provides low-current logic signals for direction and enable/PWM control.

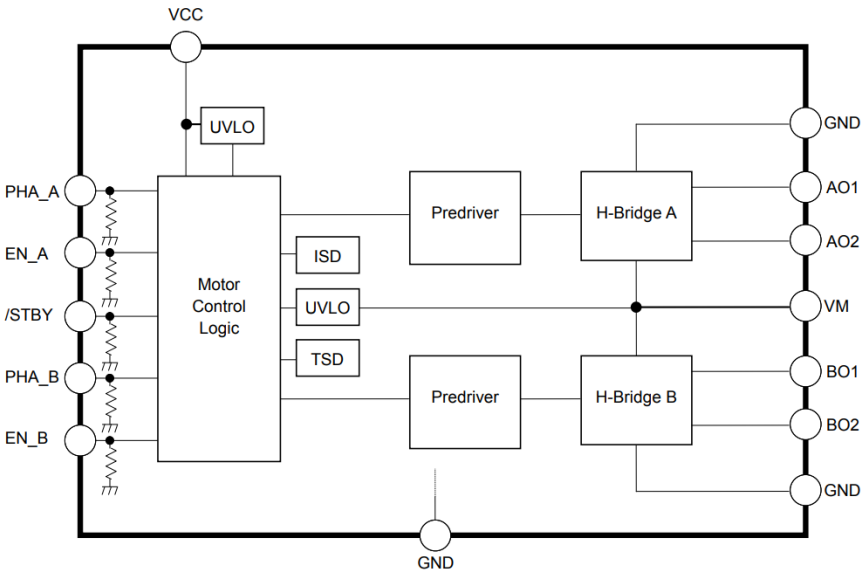


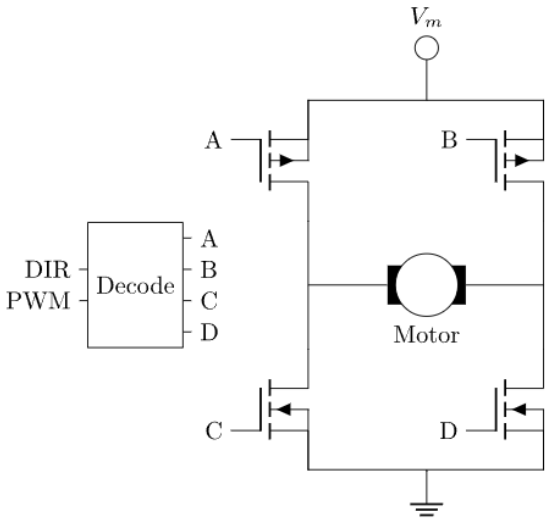
Figure 8: TC78H620FNG dual H-bridge block diagram.

The relevant TC78H620FNG functions are: dual H-bridge outputs for two brushed DC motors, direction inputs for polarity reversal, enable/PWM inputs for speed control, under-voltage lockout, over-current/short-circuit protection, and thermal shutdown. These features

made the device suitable for the fallback design because it provided the required motor-control functions while adding electrical protection against common integration faults.

1.3.2 Direction and PWM speed control

The motor driver uses an H-bridge to control both motor direction and speed. The direction input selects the conducting switch pair and therefore sets the sign of the voltage across the motor. The enable input is driven by a PWM signal, so the motor supply is switched rapidly on and off. Because the PWM frequency is much higher than the mechanical response of the motor and wheel, the rotor responds mainly to the average applied voltage.



DIR	PWM	A	B	C	D	Motor
x	0	V_m	V_m	V_m	V_m	off
0	1	0V	V_m	0V	V_m	forwards
1	1	V_m	0V	V_m	0V	reverse

Figure 9: Simplified H-bridge decoder and motor-drive truth table.

For a PWM duty cycle D, the approximate average motor voltage is

$$V_{\text{motor,avg}} \approx DV_{\text{M}},$$

where V_{M} is the motor supply voltage. This gives efficient digital speed control because the driver transistors operate primarily in low-loss on/off states rather than dissipating power continuously as a linear regulator would.

PWM behaviour was tested using the oscilloscope connections shown in Figure 10. The waveforms in Figure 11 confirm that the motor-driver output changes with the commanded duty cycle. This validates the electrical control path from the microcontroller PWM output to the motor-driver output. The remaining mechanical validation is to relate duty cycle to wheel speed, displacement, current draw, and alignment performance.

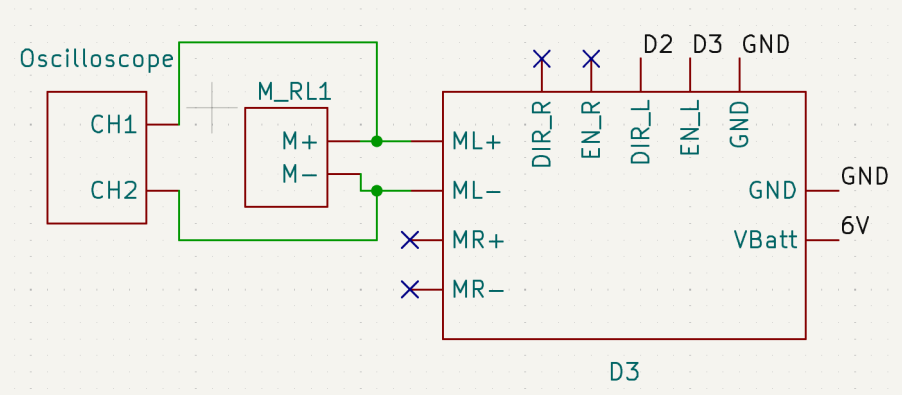
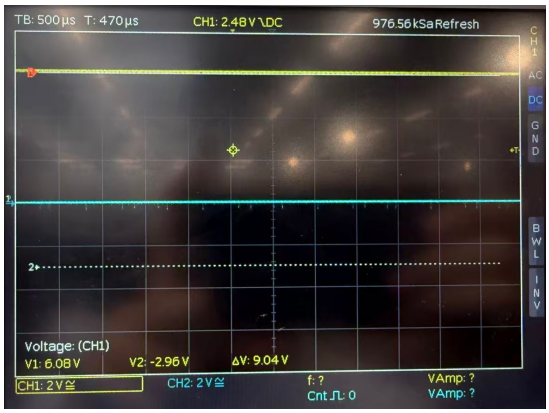
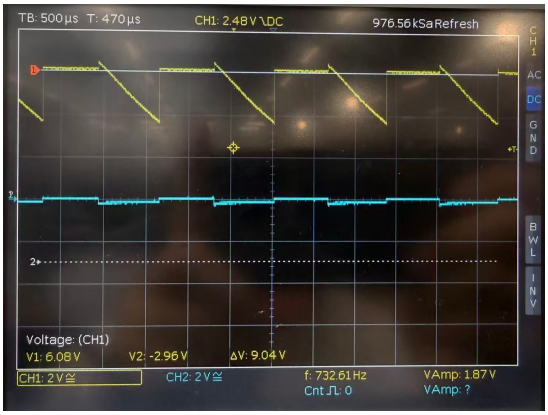


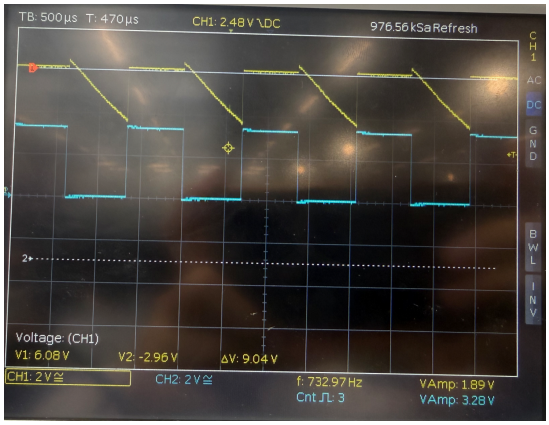
Figure 10: PWM test connections used to observe motor-driver output waveforms.



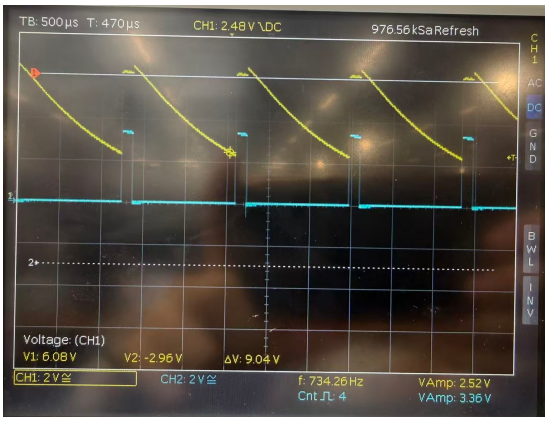
(a) PWM = 100 %, CH2 reads M-.



(b) PWM = 50 %, CH2 reads M-.



(c) PWM = 50 %, CH2 reads D3.



(d) PWM = 10 %, CH2 reads D3.

Figure 11: Oscilloscope waveforms confirming PWM switching behaviour at different duty cycles.

1.3.3 Power regulation and hardware safety

The motor power stage was designed around the TC78H620FNG operating limits. The motor supply input V_M is powered from the nominal 6 V motor rail, which is within the driver operating range of 2.5–15 V and below the 18 V absolute maximum rating. The logic supply V_{CC} is operated within the allowed 2.7–5.5 V range, making it compatible with the 3.3 V logic signals generated by the Adafruit microcontroller.

The selected driver has a maximum operating output current of 0.8 A per channel and an absolute maximum output current rating of 1.0 A. The highest load is expected during motor

startup, sudden reversal, lateral mecanum motion, or blocked-wheel conditions. The design therefore aims to keep sustained current below the operating limit and not rely on protection circuits during normal operation.

Condition	Measured current per motor	Driver limit	Margin
No-load motor spinning	0.135 A	0.8 A operating	0.665 A
Straight driving on arena surface	0.123 A	0.8 A operating	0.677 A
Lateral mecanum motion	0.121 A	0.8 A operating	0.679 A
In-place rotation	0.121 A	0.8 A operating	0.679 A
Blocked-wheel or stall check	0.180 A	1.0 A absolute max	0.820 A

Table 6: Motor-driver current-margin validation table. Readings using average out of 10 measurements. The blocked-wheel row should use the highest repeatable measured current.

The power system was proved to be safely regulated.

A key system-level risk was noise coupling between the DC motors and the microcontroller/sensor electronics. Motor startup, braking, and rapid direction changes cause transient current demands. If the microcontroller and sensors were powered directly from the same unregulated rail as the motors, voltage droop or switching noise could cause brownouts, resets, or corrupted sensor readings. The final architecture therefore separates the motor rail from the regulated logic rail: motor drivers are powered from the 6 V battery rail, while the Adafruit board is supplied through a regulated 5 V input and produces 3.3 V logic signals for the driver inputs. Local decoupling capacitors are placed near the motor-driver supply pins to suppress high-frequency switching noise.

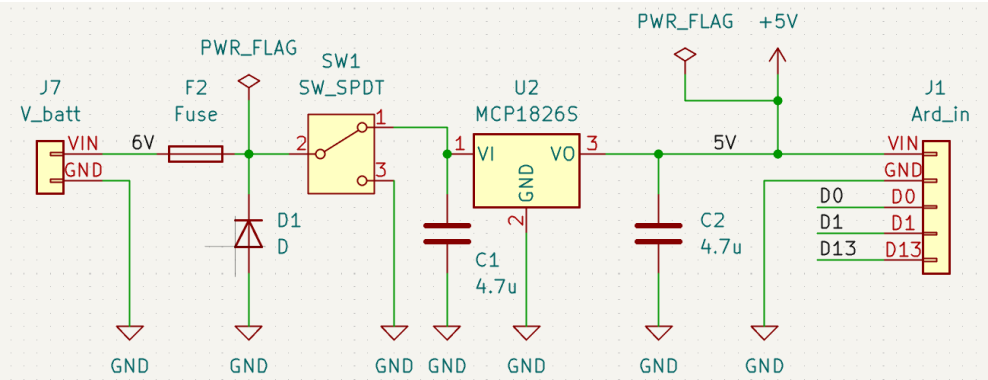


Figure 12: Power-distribution schematic showing separated motor and logic rails, regulated microcontroller input, and local decoupling.

Test	Target metric	Measured result	Pass?
Logic rail during motor startup	No reset; voltage remains within microcontroller supply tolerance.	No reset observed.	Y
Motor rail during sudden reversal	Voltage remains within driver operating range (15 V).	7 V	Y
Sensor output with motors off vs on	No classification-relevant shift.	[TBD]	[TBD]
PWM duty-cycle check	Measured duty cycle matches command within 10 %.	6%	Y
Driver temperature after continuous driving	Below safe-touch limit after 5 min.	Safe to touch	Y

Table 7: Power and noise validation tests for the motor-control subsystem.

The control system was mostly validated, meaning the design was good.

1.3.4 Motor-driver connections and pin allocation

Each TC78H620FNG controls two motors, so two driver ICs were used to control the four mecanum wheels independently. Driver D1 controls the front pair of motors and driver D2 controls the rear pair. For each motor channel, the direction pin controls polarity and the enable pin receives the PWM signal.

Pin assignments were changed during integration because several Arduino pins were already reserved by other modules, particularly the WiFi shield. Pins 5, 7, and 10 could not be used freely. The affected motor-control signals were reassigned to available digital pins: the rear-right direction signal was moved from D5 to D2, and the front-right enable signal was moved from D7 to D12. This avoided incompatible sharing between the motor drivers, WiFi shield, magnetometer, radio circuit, ultrasonic sensor, and infrared sensor.

Driver	Motor	Direction pin	PWM pin	Note
D1	M_{FL}	D8	D11	Front-left motor.
D1	M_{FR}	D13	D12	Enable reassigned from D7.
D2	M_{RL}	D4	D6	Rear-left motor.
D2	M_{RR}	D2	D3	Direction reassigned from D5.

Table 8: Final motor-driver pin allocation after resolving WiFi-shield and sensor pin conflicts.

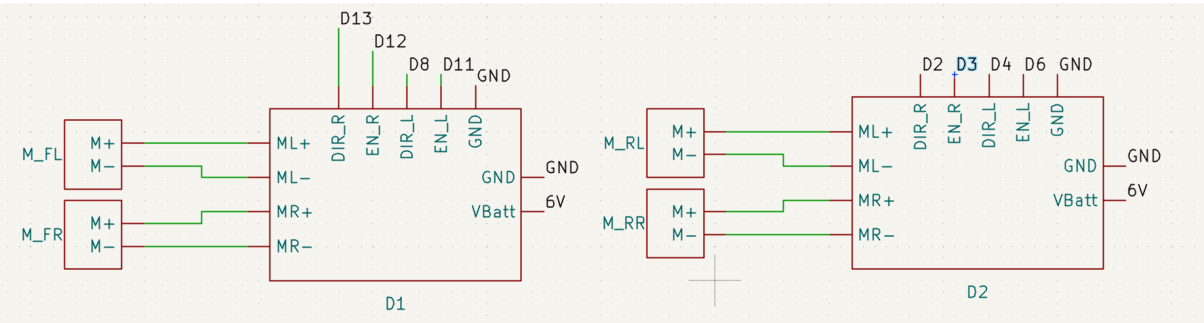


Figure 13: Motor-driver connections for the four independently controlled mecanum wheels.

1.3.5 Supply-chain delay and fallback decision

The initial motor-control plan used the WPSH207 dual-L293D motor-driver shield. This was attractive because it provided four H-bridge channels on one Arduino-compatible board, simplifying wiring, mechanical mounting, and pin management. It also included terminal blocks, separate motor and logic power connections, and integrated kickback protection. The shield was ordered on 21 May with an expected delivery date of 5 June.

A supplier check on 7 June showed that the delivery estimate had moved to 28 June. This created a schedule risk: waiting for the shield would have delayed motor integration, sensor placement, system testing, and final arena preparation. The design was therefore changed to use the available TC78H620FNG motor-driver ICs. The fallback option increased wiring and integration effort but preserved the project schedule and allowed drivetrain testing to continue.

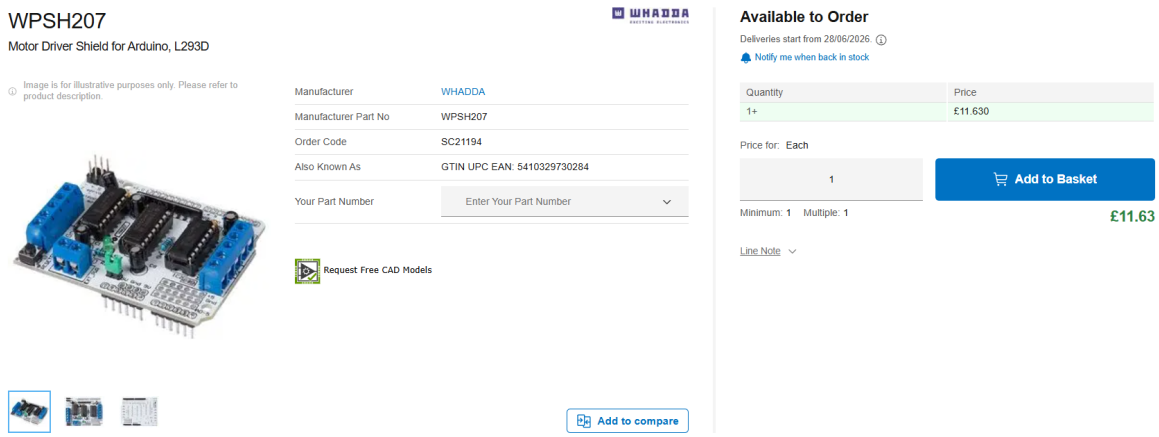


Figure 14: WPSH207 supplier page showing delayed motor-shield availability.

Criterion	WPSH207 shield	TC78H620FNG fallback
Availability	Delayed beyond useful integration window.	Available immediately.
Integration complexity	Lower: single shield, terminal blocks, fewer wires.	Higher: two ICs/modules and more custom wiring.
Schedule risk	High after delivery estimate moved to 28 June.	Lower; allowed integration to continue.
Current capability	Suitable for nominal 100 mA motors, subject to stall-current check.	Suitable for nominal operation, subject to current-margin validation.
Pin flexibility	Shield pin usage less flexible.	Pin allocation could be adapted around WiFi and sensor conflicts.
Decision	Rejected due to delivery risk.	Selected to protect integration and testing time.

Table 9: Fallback decision after the original WPSH207 motor shield became unavailable within the project schedule.

This decision is important because it demonstrates a project-management trade-off rather than a purely technical component preference. The fallback design was less convenient, but it reduced the larger risk of reaching system testing without an integrated drivetrain.

1.4 Chassis Iteration and Mechanical Design

The chassis is the structural foundation of the rover. It must hold the drivetrain rigidly, support the battery and microcontroller, maintain repeatable sensor geometry, and remain light enough for the 750 g limit. The chassis was developed iteratively in FreeCAD so that packaging, sensor placement, mass distribution, and manufacturability could be assessed before fabrication.

1.4.1 Iteration 1: single-layer chassis

The first mechanical iteration used a single-tier flat-bed chassis measuring approximately 120 mm by 150 mm. This gave a plan area of

$$A = 120\,\text{mm} \times 150\,\text{mm} = 18\,000\,\text{mm}^2 = 180\,\text{cm}^2.$$

This design was rejected after CAD evaluation for two reasons. First, once the motors and mecanum wheels were included, the total vehicle width exceeded approximately 230 mm; this reduced clearance between obstacles and was poorly matched to rocks spaced only 500 mm apart. Second, the sensitive front-end sensors were placed on a projected front extension that would later be fixed using adhesive. This introduced risks of misalignment, vibration, and detachment during movement. Because the radio coil and ultrasonic sensor are directionally sensitive, even a small change in angle could reduce measurement repeatability.

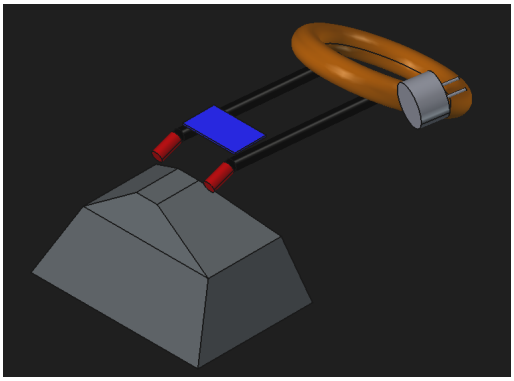


Figure 15: Initial front sensor-placement concept.

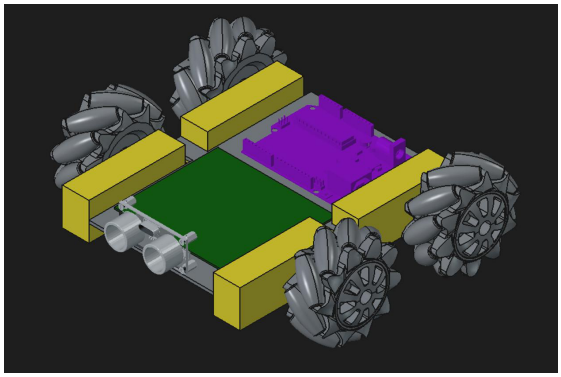
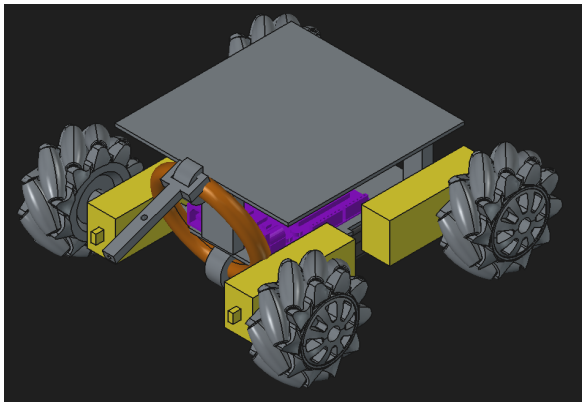


Figure 16: FreeCAD rendering of the rejected 120 mm by 150 mm single-tier chassis.

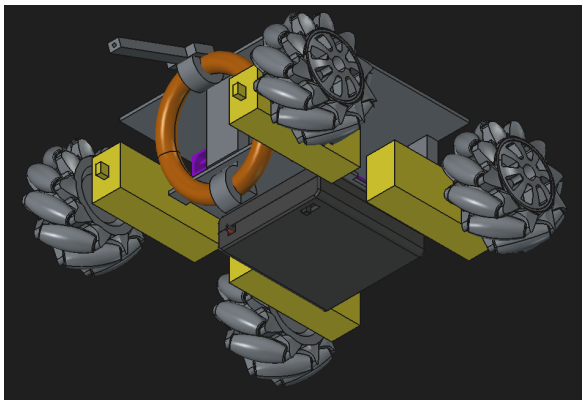
1.4.2 Final dual-layer chassis

The final design uses a compact dual-layer architecture. The lower layer carries mechanically heavy and electrically noisy components: motors, battery holder, lower structural supports, and the main control board. The upper layer carries the sensor and circuit modules. This improves physical packaging, lowers the centre of mass for drivetrain and battery components, and reduces direct mechanical/electrical coupling between the motors and the analogue sensing circuits.

The dual-layer layout also improves sensor placement. The radio coil is mounted at the front of the rover using custom 10 mm internal-diameter clamps, allowing it to approach a target without the main chassis contacting the rock first. The same design principle was used for the other directional sensors: mounting points were treated as functional geometry, not simply as convenient spare space.



(a) Front/isometric view.



(b) Rear/isometric view.

Figure 17: Two isometric views of the final dual-layer chassis design.

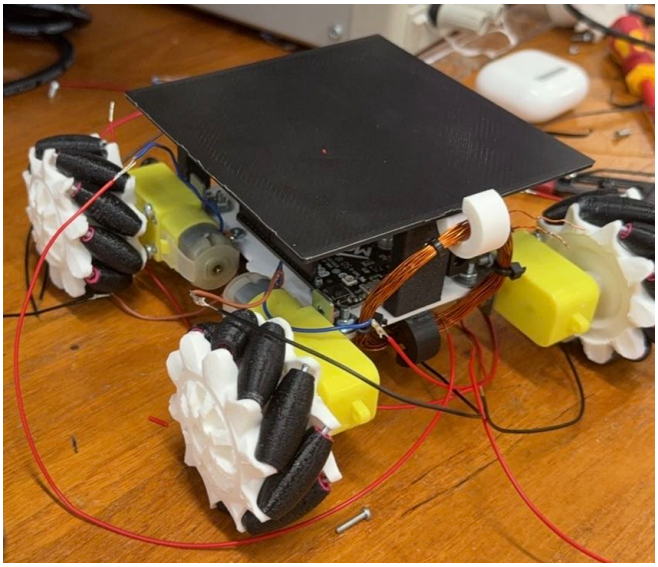
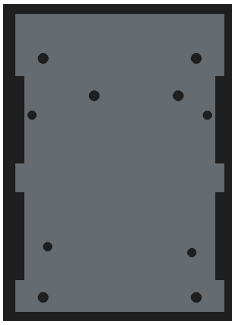


Figure 18: Final chassis assembly showing the two-layer mechanical layout.

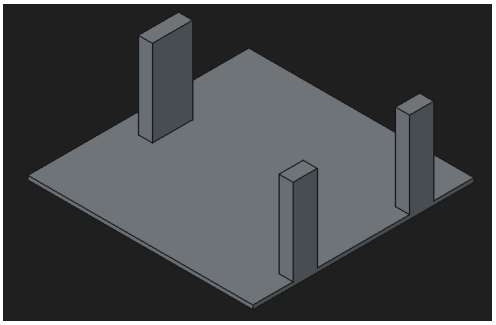
1.4.3 Models, dimensions, and mass allocation

The main fabricated chassis components are shown in Figure 19. Their dimensions were selected to minimise footprint while preserving mounting space for the motors, battery, Adafruit board, WiFi shield, driver circuits, and sensors.

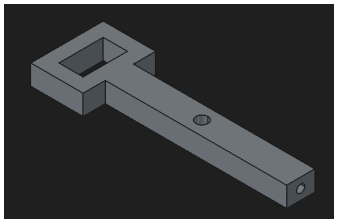
- **Bottom layer, 70 mm by 100 mm:** supports the drivetrain and power system. The four 18 mm by 70 mm DC motors are fixed using right-angle brackets. Locating motors and batteries low in the chassis reduces centre of mass and improves stability during lateral mecanum movement.
- **Top layer, 112 mm by 110 mm:** supports the sensor and circuit modules. The larger top area provides routing and mounting space while keeping the lower mechanical footprint compact.
- **Sensor clamps and extension bracket:** provide a rigid front mounting structure for the radio coil and other sensing hardware, reducing the chance of geometry changes after repeated driving trials.



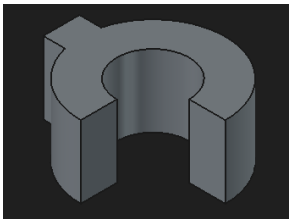
(a) Bottom layer.



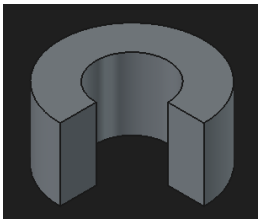
(b) Top layer with 45 mm supports.



(c) Extension bracket.



(d) Bottom clamp, $\oslash = 10$ mm.



(e) Top clamp, $\oslash = 10$ mm.

Figure 19: CAD views of the main fabricated chassis components.

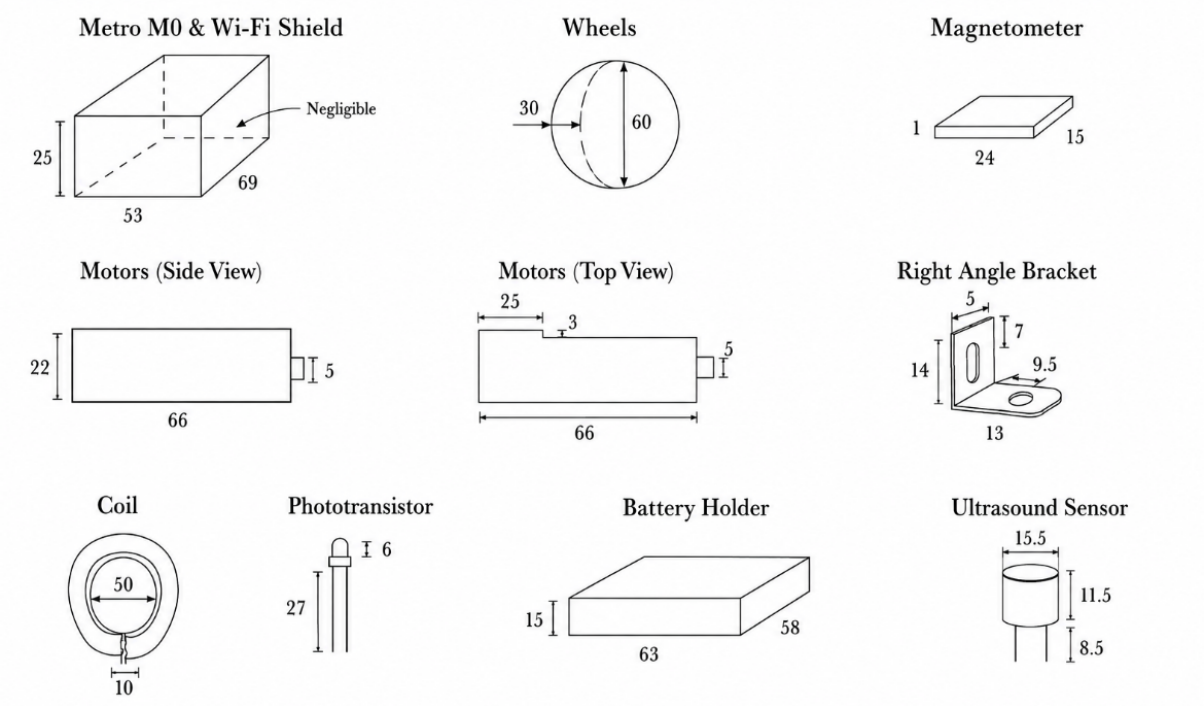


Figure 20: Dimensions of the main mechanical and electronic components used for chassis packaging.

Table 10 shows the measured mass of the major known components. The subtotal is 489.9 g, leaving 260.1 g for the chassis, fasteners, motor drivers, circuit boards, wiring, adhesives, and protective mounts before reaching the 750 g limit. This margin is useful but not large enough to rely on estimates; the final assembled mass must be measured after all wiring and mounting hardware are installed.

Component	Mass / g
Sensors	14.6
Motors	119.8
Wheels	197.2
Adafruit board	20.7
WiFi shield	18.6
Batteries	$26.1 \times 4 = 104.4$
Battery holder	14.6
Measured subtotal	**489.9**
Remaining allowance before 750 g limit	**260.1**

Table 10: Mass budget for the main known components, excluding chassis, wiring, fasteners, motor-driver boards, and final mounting hardware.

Remaining item	Measured mass / g
3D-printed bottom layer	15.1
3D-printed top layer and supports	39.2
Sensor clamps and extension bracket	[TBD]
Motor-driver boards or IC carrier boards	[TBD]
PCB/stripboard and soldered components	[TBD]
Wires, connectors, tape, and heat-shrink	[TBD]
Screws, nuts, brackets, and adhesive	[TBD]
Final assembled mass	**[TBD]**

Table 11: Remaining mass budget required to confirm compliance with the 750 g limit.

1.5 Hardware Prototyping Evolution: Breadboard, Stripboard, and PCB

The sensor and power circuits were also iterated mechanically because the physical build medium affected reliability, mass, footprint, and signal integrity. The progression from breadboard to stripboard to PCB was therefore a system-integration decision, not just a manufacturing preference.

1.5.1 Breadboard prototype

The initial integration phase used solderless breadboards because they allowed rapid proof-of-concept testing. Component values could be changed quickly while tuning op-amp gains and checking sensing range. During this stage, the radio circuit was observed to detect a rock at approximately 4.5–5 cm before approaching the 3.3 V rail limit.

Breadboards were rejected for the final rover because they were mechanically unsuitable for a moving mecanum platform. Loose jumper wires and weak contact retention would make intermittent disconnections likely during vibration, rotation, and obstacle contact. Breadboards also imposed a mass and footprint penalty: each board weighed approximately 50 g, and multiple boards plus jumper wires would consume too much of the remaining mass and top-layer area.

1.5.2 Stripboard compromise

Stripboard was evaluated as an intermediate solution between temporary breadboards and a permanent PCB. The layout was planned using Stripboard-Editor, which helped identify

connected nodes and track conflicts. The layout required the sensitive radio and infrared sensor front ends to be close to the front of the board, with 3.3 V, 5 V, and ground rails routed near the relevant subsystems and the regulator isolated toward the rear.

The stripboard approach was abandoned because it did not scale cleanly to the full multi-stage sensor suite. The number of required links caused overlapping wire routes and a board footprint larger than the 112 mm by 110 mm top-layer space. It also retained a non-negligible risk of wiring faults and rework, especially where analogue sensing, power regulation, and motor-control signals had to coexist on the same moving platform.

1.5.3 Custom PCB solution

The final hardware integration was consolidated onto a custom PCB designed in KiCad. This decision improved the system in three ways. First, it reduced mass by replacing bulky breadboards and long jumper wires with a single FR4 substrate and routed copper traces. Second, it improved mechanical robustness because soldered joints and a rigid board are less vulnerable to vibration than breadboard jumpers. Third, it improved signal integrity by shortening analogue signal paths and reducing parasitic breadboard effects on sensitive 89 kHz radio and 40 kHz ultrasound circuitry.

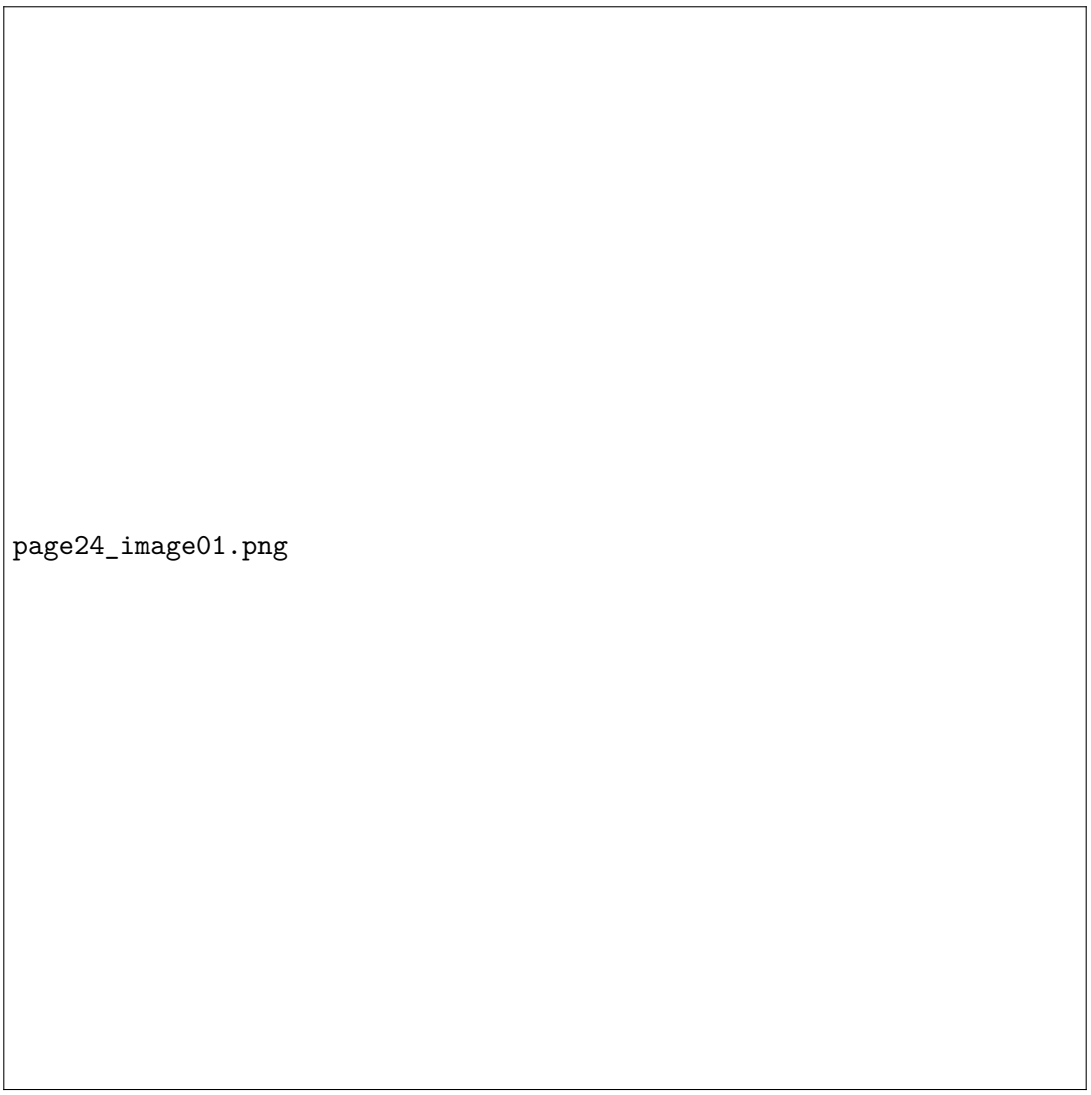


Figure 21: Hardware evolution from the initial heavily wired breadboard prototype to the final custom PCB design.

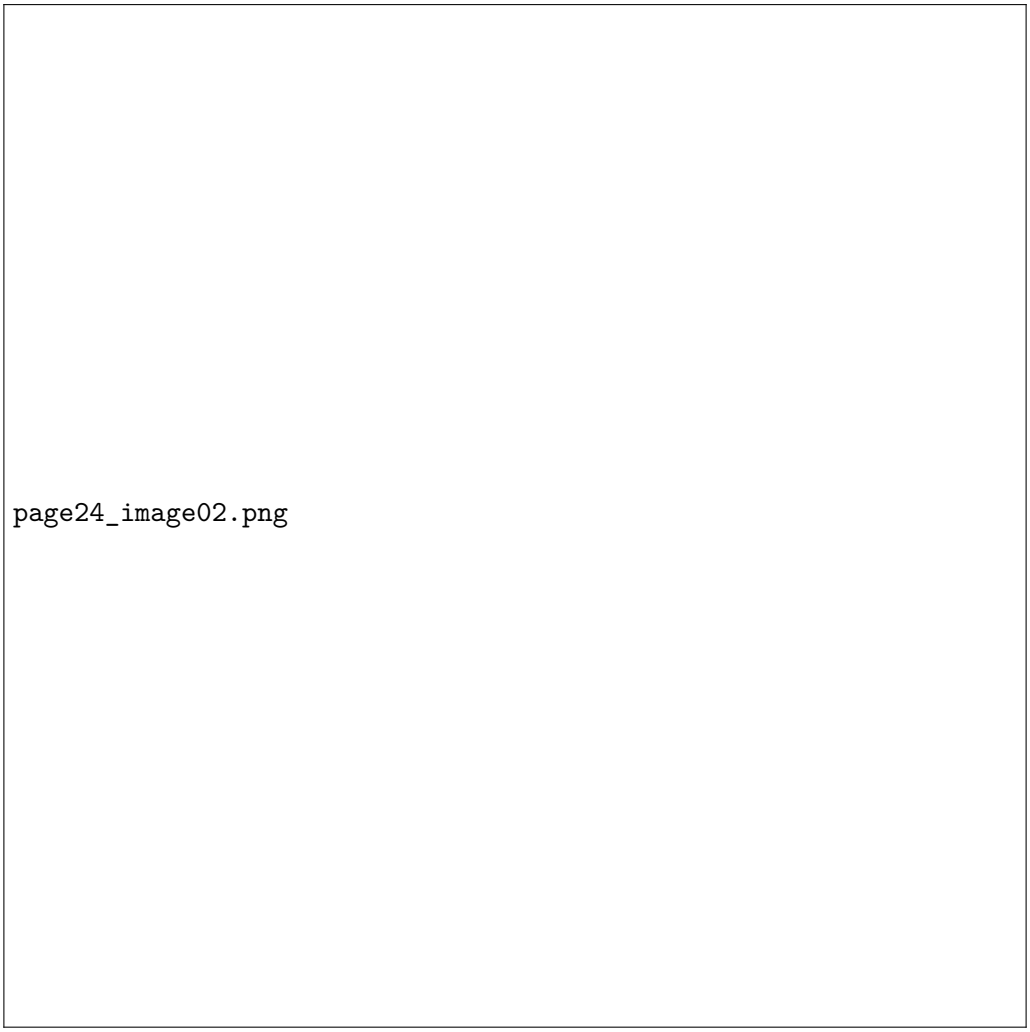


Figure 22: PCB layout used to consolidate the rover's sensing and support electronics.

Medium	Strength	Limitation	Decision
Breadboard	Fastest for gain tuning and early signal checks.	Heavy, bulky, and unreliable under vibration.	Used only for early proof-of-concept.
Stripboard	More permanent than breadboard and can be hand-built quickly.	Routing became congested and exceeded available top-layer area.	Abandoned after layout evaluation.
Custom PCB	Lowest wiring complexity, best mechanical robustness, cleaner signal routing.	Requires design time, fabrication risk, and less flexibility after manufacture.	Selected for final integration.

Table 12: Comparison of physical prototyping media for final rover hardware integration.

1.6 Links to Evidence

Link to holonomic movement sequence